



Krishi AgriTech Platform

Farmer & Wheat Crop Monitoring System — End-to-End Workflow & Architecture Guide

NEXT.JS 16 APP ROUTER

TYPESCRIPT

REDUX TOOLKIT

LEAFLET SATELLITE GIS

RABI 2025-26 SEASON (MVP)

1. Executive Overview & Purpose

Krishi AgriTech is an integrated digital agronomy platform designed to digitize farm parcel boundaries, track chronological wheat phenology, empower farmers with real-time micro-climate intelligence, and provide central management with satellite-driven ground truth visibility.



Farmer Portal (/farmer)

Self-service mobile/web portal for daily activity logging (Date/Time), spray advisories, disease detection, and live mandi rates.



Admin Center (/admin)

GIS Command Center with interactive satellite map, farmer parcel fly-to zoom, enrollment verification, and activity feeds.



Field Officer (/field-officer)

Ground auditing interface for cadastral boundary validation, pest threshold checks, and agronomist feedback.

2. Step-by-Step Operational Lifecycle Workflow

1

Farmer Enrollment & KYC Verification (/admin/farmers)

Central administration registers the grower's legal identity (Aadhaar/CNIC, Contact, Village, and Total Landholding). The grower is assigned an official code (e.g. FARM-2026-001 for Sardar Gurpreet Singh).

2

Satellite Land Parcel Geofencing (/admin/land-parcels)

Using the interactive **Satellite Location Picker**, officers drop a draggable GPS pin or use live geolocation to map exact cadastral coordinates (e.g. Khasra #412/1, 18.0 Acres) with real-time acreage polygon computation.

3

Crop Cycle Initiation & Phenology Monitoring (/admin/crop-cycles)

The wheat cycle (HD-2967) is tracked across 11 agronomic stages (Sowing → CRI → Tillering → Jointing → Booting → Heading/Flowering → Maturity). Sentinel-2 NDVI canopy vigour (0.86 Optimal) is continuously monitored.

4

Daily Field Operations & Activity Logging (/farmer)

Growers open the "+ Record Activity" modal with exact **Date & Time** inputs to log operations like Urea Top-Dressing (1 Bag/Acre = 18 Bags (900 Kg) Total, ₹4,800 cost) or Canal Irrigation, instantly syncing to the central audit feed.

5

Micro-Climate Weather & Smart Spray Matrix (/farmer)

Live agro-meteorological telemetry displays temperature (27°C), humidity (55%), root-zone moisture (64%), and calculates a **Safe Spray Window (07:00 AM – 11:30 AM)** based on wind velocity (<10 km/h) and 0% rain chance.

6

AI Crop Doctor & Pest Early Warning

Growers capture leaf images to run on-device computer vision checks for Stripe/Yellow Rust (Puccinia striiformis) or connect directly with the Punjab Agromet Helpline (0800-15000).

7

Harvest Settlement & Mandi Procurement (/admin/harvest)

Yield estimation (54.5 Mnds/Acre) is verified at harvest. Live grain pricing compares Govt MSP (₹2,275/Qtl) vs Khanna Mandi (₹2,420/Qtl) vs Silo Gate Rates (₹2,480/Qtl for <11% moisture).

3. Data Entities & Architecture

ENTITY	KEY ATTRIBUTES	DESCRIPTION & ROLE
Farmer	id, fullName, cnic, phone, village, totalLandAcres	Verified grower identity, landholding summary, contact records.
LandParcel	id, farmerId, parcelCode, khasraNo, acreage, lat, lng	Cadastral plot geometry with satellite overlay coordinates.
CropCycle	id, cropVariety, season, currentStage, ndviScore, yield	Lifecycle progress from sowing to harvest with canopy vigour.
FieldActivity	id, farmerId, activityType, executedDate, dosage, cost	Timestamped log of fertilizers (Urea/DAP), water, and chemicals.
WeatherDay	date, tempMax, tempMin, windSpeed, rainChance, advice	Micro-climate forecasts and automated agronomic recommendations.

4. Application Route Architecture



Admin Routes

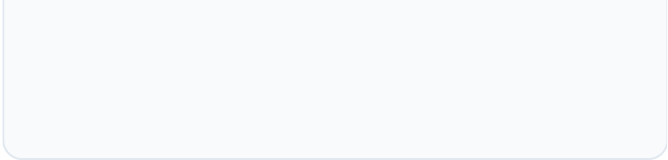
- /admin: Main Command Center + GIS Map + Farmer Dropdown
- /admin/farmers: Farmer Enrollment & Records
- /admin/land-parcels: Satellite Location Picker & CAD Plots
- /admin/crop-cycles: Wheat Phenology & NDVI Tracking
- /admin/activities: Live Field Activity Log Feed



Farmer & Officer Routes

- /farmer (Tab 1): My Wheat Crop & Phenology Stages
- /farmer (Tab 2): Land Parcels & Satellite Map
- /farmer (Tab 3): Field Operations Log & Date/Time Modal
- /farmer (Tab 4): AI Leaf Disease Doctor
- /farmer (Tab 5): 24-hr Weather & Live Mandi Rates
- /field-officer: Ground Scouting & GPS Validation

- /admin/harvest: Silo Gate Intake & Yield Settlement
- /admin/reports: Farm Analytics & Audit Logs



💡 **Key MVP Optimization:** The Farmer Portal was streamlined to eliminate the legacy Digital Passbook module, focusing exclusively on active crop stages, daily fertilizer/irrigation logging, weather forecasting, and market pricing.